

SWAYAM KOLABKAR

Data / AI / ML Engineer

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SUMMARY

AI/ML Engineer with over 3 years of experience designing and deploying scalable machine learning systems, data pipelines, and cloud-native architectures across AWS, Azure, and GCP. Demonstrated expertise in fine-tuning large language models (LLMs) such as BERT, GPT, Phi, and Qwen for applications in NLP, predictive analytics, and real-time decisioning. Skilled in building distributed ETL workflows, deploying ML models in production environments, and developing knowledge graphs for complex data relationships. Strong background in data warehousing, API integration, and MLOps practices, with a focus on delivering intelligent, efficient, and business-aligned solutions.

EXPERIENCE

AI/ML Engineer

March 2025 – Present

Topclub

USA

- Fine-tuned AI models like **Phi**, **Qwen**, **BERT**, and **GPT** to boost predictive analytics and user engagement in **soccer** platforms.
- Deployed and monitored **ML models** via **SageMaker** and **Azure ML** for scalable, **low-latency** inference in live soccer analytics and performance systems.
- Built distributed workflows using **Databricks**, **AWS Lambda**, and **SSIS** for real-time and batch soccer data processing, improving system reliability by **25%** and migration efficiency by **40%**.
- Developed **graph-based** knowledge representations with **Neo4j** and **AWS Neptune** to enable entity linking, multilingual context understanding, and semantic querying within datasets.
- Designed and optimized **Star** and **Snowflake** schemas to support advanced reporting and analytics, while revamping **ETL** workflows to improve operational efficiency by **25%**.
- Unified data from **NetSuite** and **Salesforce** to support real-time **NLP inference** and reporting, while collaborating on **LLM** tuning and prompt optimization using **RAG** pipelines.

Machine Learning Engineer

Jan 2024 – Feb 2025

Virtusa

USA

- Architected and implemented scalable, **fault-tolerant** data pipelines with **Hadoop**, **Apache Spark**, and **Airflow**, cutting **ETL** processing time by **40%** and enhancing throughput for critical workloads.
- Containerized data services with **Docker** and automated **CI/CD** pipelines using **AWS Lambda**, reducing deployment time by **30%**, increasing success rate by **15%**, and enhancing the reliability of ER production systems.
- Developed and fine-tuned **Talend** jobs for extracting, transforming, and batch loading multi-source data, enhancing data architecture management and increasing migration efficiency by **30%**.
- Engineered end-to-end data workflows using **Snowflake** for data transformation, encompassing preprocessing, feature extraction, data modeling, and deployment to support analytics and real-time **ML inference**.
- Built a **multi-terabyte**, end-to-end Data Warehouse on **Amazon Redshift** to handle millions of daily records, increasing data processing capacity by **50%** and enhancing query performance by **35%**.
- Developed **RESTful APIs** with **Spring Boot** to facilitate patient-provider data exchange, boosting Emergency Room responsiveness by 200% through Agile methodologies and real-time data simulators, resulting in reduced technician response times and improved access to patient information.

Machine Learning Engineer

Jan 2020 – July 2021

Experion Technologies

India

- Designed and implemented scalable data pipelines using **AWS (S3, EC2, RDS, Lambda, Redshift)** and **Azure Data Factory**, supporting real-time analytics and lowering data processing costs by **30%**.
- Developed automated **ETL workflows** with **Python**, **Airflow**, and **PySpark**, integrating data lineage tracking and validation scripts to reduce manual QA time by **50%** and decrease data processing errors by **25%**.
- Implemented **Apache Kafka** for real-time data ingestion and processing, supporting event-driven architecture and enhancing data freshness across reporting layers.
- Designed and enhanced **Power BI** dashboards and data models to empower stakeholders with faster insights, significantly boosting data-driven decision-making.
- Developed ingestion pipelines using **Neo4j** and **AWS Neptune** to process and analyze large interconnected datasets, enabling graph analytics for advanced **business intelligence applications**.

PROJECTS

BOSTON AIRNUB ANALYTICS & PRICE CORRELATION | Tableau | Python | Web Scraping | PostgreSQL

- Developed a Python-based visualization project to analyze the impact of various features (like area, location, bedrooms) on house prices. Created correlation matrices to identify key influencing factors and visualized trends to support data-driven insights. Cleaned and transformed raw data, including numeric and non-numeric fields, to improve analysis accuracy and decision-making.

POKEDEX FOR STARTER POKÉMONS Python | Pandas | NumPy | Seaborn | Tableau | GCS | SQL

- Designed an interactive Tableau dashboard to visualize and compare starter Pokémon stats using user-centric design. Focused on building a playful yet informative data storytelling experience through intuitive visuals.

GITHUB PROJECTS – DATA CLEANING & CODING SQL | PostgreSQL | MySQL | Python | R (tidyverse) | GitHub | Bash

- Compiled a series of hands-on projects focused on data cleaning, manipulation, and visualization using SQL, Python, and R. Notable work includes a Data Cleaning project using SQL that improved data quality through structured queries and transformations. Demonstrated strong foundational skills in handling raw datasets, optimizing queries, and preparing data for analysis.

SKILLS

Methodologies: Agile, Scrum, JIRA

Languages: Python, Java, C++, CSS, HTML, PL/SQL, Shell Scripting

Big Data Tools: Hadoop, MapReduce, Hive, Apache Spark, Pig

ETL and Workflow Tools: SSIS, Apache NiFi, Apache Kafka, Talend, Apache Airflow, Postman

Cloud Platforms and Data Warehousing: AWS (S3, EC2, RDS, Lambda, Redshift), Azure, Snowflake, GCP, Big Query

Containerization and CI/CD Tools: Docker, Kubernetes, Jenkins, GitLab

AI and ML Technologies: BERT, GPT, LSTM, CNN, KNN, MLE, PCA, RL, SVM, Decision Trees, Random Forests, Regression

Framework/Libraries: React.js, Matplotlib, NumPy, PyTorch, Pandas, SciPy, Sklearn, Seaborn, TensorFlow, Springboot

Business Infrastructure and Visualization Tools: Tableau, Power BI, SSRS, Looker, NFS, NGINX, Samba, VMware

Databases: MongoDB, MySQL, DBeaver, DataGrip, JDBC, PostgreSQL, Neo4j, SQL Server

Version Control Systems: Git, GitHub, AWS CodeCommit, Apache subversion

Integrated Development Environments (IDEs): PyCharm, Jupyter Notebook, IntelliJ, Visual Studio

EDUCATION

Master of Science in Information Systems

Stevens Institute of Technology

August 2021 – May 2023

Hoboken, NJ

Relevant Coursework: AI, ML, NLP, Big Data Analytics, Database Systems, Operating Systems, Linux, Deep Learning, Statistics

Bachelor of Engineering in Computer Science

University of Mumbai

July 2016– October 2020

Mumbai, India

CERTIFICATIONS

- Google Data Analytics
- Visualizing Citibike Trips with Tableau
- JPMorgan Chase Software Engineering Virtual Experience